2025-2026 Biomolecular Engineering and Bioinformatics: Biomolecular

Mathematics

•MATH 3 or Math Placement
MATH 19A
Calculus I
[F/W/Sp]

•MATH 19A
MATH 19B
Calculus II
[F/W/Sp]

•MATH 3 or Math Placement
AM 10 or Math 21
Linear Algebra
[F/W/Sp]

•MATH 19B (AM 30 or Math 23B recommended)
STAT 131
Intro to Probability Theory
[F/W/Sp]

Chemistry

Select one of the following General Chemistry series:

•Math 2 or math placement of 200 or higher
CHEM 3A
General Chemistry
[F/W/Sp]

•Chem 3A
CHEM 3B/BL
General Chemistry
[F/W/Sp]

•Chem 3B
CHEM 3C/CL
General Chemistry
[F/W/Sp]

OR

•Chem 4 Prep ALEKS module. Math 3 or Math Placement.
CHEM 4A/AL
Advanced General Chemistry
Not offered 2025-26

•Chem 4A
CHEM 4B/BL
Advanced General Chemistry
Not offered 2025-26

Biology, and Biochemistry

• CHEM 3A, or CHEM 4A
BIOL 20A
Cell and Molecular Biology
[F/W/Sp]

•BIOL 20A
BME 105
(Strongly Recommended)
Genetics in the Genomics Era
[W/Sp]

OR

•BIOL 20A and BIOE 20B
BIOL 105
Genetics [F/W/Sp]

Select one of the following Biochemistry series:

•CHEM 8B and BIOL 20A
BIOC 100A
Biochemistry and Molecular Biology [F]
&
•BIOC 100A
BIOC 100B
Biochemistry and Molecular Biology [W]

•BIOL 20A
BME 101
Molecular Biology for Biomolecular Engineers [F]
&
•Chem 8B
CHEM 103
Biochemistry
[S]

•BIOL 20A
BME 101
Molecular Biology for Biomolecular Engineers [F]
&
•BIOL 20A and BIOE 20B, and CHEM 8B
BIOL 100
Biochemistry
[F/WSp]

Physics

•MATH 19A, corequisite of Phys 5L AND Math 19B
PHYS 5A/L
Introduction to Physics I
[F/W/Sp]
OR
PHYS 15A/ 5L
Honors Introduction to Physics I [F]

•PHYS 5A/L and MATH 19B; concurrent enrollment in PHYS 5M
PHYS 5B/M
Introduction to Physics II
[W/Sp]

Laboratory Courses

•CHEM 1M or 3BL or 4BL
BME 21L
Intro. To Basic Laboratory Techniques
[F/W]

•BME 21L
BME 22L
Foundations of Design and Experimentation in Molecular Biology
[W/Sp]

Organic Chemistry

•CHEM 3C or CHEM 4B
CHEM 8A
Organic Chemistry
[F/W]

•CHEM 8A
CHEM 8B
Organic Chemistry
[W/Sp]

Bioinformatics and Bioethics

BME 80G
Bioethics in the 21st Century: Science, Business, and Society
[Sp]

•BIOL 20A
BME 160 (6 units)$^{\Omega}$
Research Programming in the Life Sciences
[F/W]

•BME 105 or BIOL 105 or BIOC 100A or CHEM 103 or declared BMEB majors
BME 110
Computational Biology Tools
[F/W/Sp]

•ELWR and BIOL 20A
BME 185 (DC)
Technical Writing for Biomolecular Engineers
[F/W/Sp]

•BME 160 or BME 205
BME 163
Applied Visualization and Analysis of Scientific Data
[Sp]

Modeling & Design Sequence

Choose one of the following sequences

•BIOL 20A
BME 177
Engineering Stem Cells
[Sp]
&
•BME 22L
BME 177L (2 units)
Engineering Stem Cell Lab
[Sp]

• BIOL 100 or BIOC 100A or CHEM 103
BME 128
Protein Engineering
[W]
&
•BME 22L
BME 128L (2 units)
Protein Engineering Lab
[W]

•BIOL 105 or BME 105 or METX 140
BME 130
Genomes [F]
&
•BME 22L
BME 123L (2 units)
Long Read Sequencing
[F]

Elective: *One of the following (course used as an Elective cannot be used to satisfy other major requirements)*

AM 115, AM 147, BIOL 115*, METX 100, METX 140, BIOC 100C, BME 118, BME 122H, BME 123L, BME 128, BME 128L, BME 130, BME 132, BME 140, BME 175, BME 177, BME 177L, BME 178*, ECE 104, or any 5-credit biomolecular engineering graduate course (BME 201-279)

*Courses have additional prerequisites not covered by the major requirements

Biomolecular Capstone: *Students must complete one of the following:*

Bioinformatics Capstone#	iGEM	Senior Design	Senior Thesisα
•BME 160, STAT 131, and prev. or conc. enrollment in Biochemistry **BME 205** Bioinformatics Models and Algorithms [F]	•prev or conc. enrollment in BME 185 or CSE 185E **BME 180(2 units)** Professional Practice in Bioengineering [W]	•BME 22L; BIOL 100 or BIOC 100A or CHEM 103 or BME 101; Previous or concurrent enrollment in BME 185 **BME 129A** Biomolecular Engineering Project I [F]	**BME 195** Senior Thesis Research [F]
•BME 205 **BME 230A** Introduction to Computational Genomics and Systems Biology [W]	•BME 180 and instructor permission **BME 188A(2 units)** Synthetic Biology – Mentored Research A [Sp]	•BME 129A **BME 129B** Biomolecular Engineering Project II [W]	**BME 195** Senior Thesis Research [W]
•BME 129A and BME 129B **BME 129C** Biomolecular Engineering Project III [Sp]	•BME 188A and instructor permission **BME 188B** Synthetic Biology – Mentored Research B [Su]	•BME 129B or BME 230A **BME 129C** Biomolecular Engineering Project III [Sp]	**BME 195** Senior Thesis Research [Sp]
	•BME 188B and instructor permission **BME 188C** Synthetic Biology – Mentored Research C [Su]		

2025-2026 Biomolecular Engineering and Bioinformatics: Biomolecular

______	______	Spring ______	______
Fall	Winter		Summer

______	______	Spring ______	______
Fall	Winter		Summer

______	______	Spring _____	______
Fall	Winter		Summer

______	______	Spring ______	______
Fall	Winter		Summer

Legend

Ω Students with no prior programming experience are advised to take CSE 20 prior to BME 160

The Bioinformatics capstone is programming heavy. Students interested in this capstone are advised to take additional programming classes.

α The thesis option consists of 15 credits of Senior Thesis Research (BME 195) in Biomolecular Engineering. Students pursuing the senior thesis option must write a two-page thesis proposal and seek approval of their project from the undergraduate director in the quarter preceding the independent study courses, typically spring quarter of the third year. Students spend three or more quarters working on their thesis projects. Students should plan on 15 units of BME 195 split over 3 quarters.

Due to course overlap between the biomolecular engineering and bioinformatics (BMEB) B.S., and the bioinformatics minor, none of these double major or major/minor combinations will be considered. Other major/minor combinations are permitted and encouraged. Double majors with the biotechnology B.A. and majors in the Humanities. Social Sciences, and Arts Divisions are specifically encouraged.

Exit Requirements

Students are required to submit a portfolio, exit survey, and attend an exit interview. The portfolios must be turned in by the last day of the quarter of graduation, and will be reviewed quarterly by the undergraduate director. Exit interviews are scheduled during the last week of the quarter by Baskin Engineering advising office, generally as small group interviews. Additional information can be found in the program catalog statement.

1. Portfolio
2. Exit Survey
3. Exit Interview